The empirical closed form for OEIS A209533, proved

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A209533 carries an empirical closed form for $a(n)$ — not a recurrence relating consecutive terms but an explicit formula. It is true, and it is decidable rather than empirical. The entry counts arrays subject to a condition on every 2×2 subblock, hence on two consecutive lines, so the lines of an array are the vertices of a finite digraph, an array is a walk in it, and $a(n)$ is a walk count on $S = 6561$ vertices. Any function of the form $\sum_j p_j(n)\lambda_j^n$ satisfies the monic linear recurrence whose characteristic polynomial is $\prod_j (x - \lambda_j)^{\deg p_j + 1}$; here that polynomial is $q(x) = (x - 16)(x - 1)$, of degree 2. Two things are then checked exactly: that the walk count satisfies the same recurrence from some index on, which the Cayley–Hamilton theorem reduces to a finite computation, and that the two sequences agree at 2 consecutive indices past that point. A monic recurrence determines every later term from its predecessors, so agreement there is agreement for good.

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1 The conjecture

OEIS A209533 is "Half the number of (n+1)X8 0..2 arrays with every 2X2 subblock having exactly two distinct clockwise edge differences.". It has offset 1 and begins

$$257, 4097, 65537, 1048577, 16777217, 268435457, 4294967297, 68719476737, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: a(n) = 16^(n+1)+1

As of the "Last modified" line on the live entry (Oct 03 2025 15:19:13, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved. Write

$$f(n) \ = \ 16^{n+1} + 1.$$

2 The count is a walk count

The entry's defining condition is local: it is a condition on every 2×2 subblock, hence on two consecutive lines. Take as vertices the admissible configurations of such a window and put an edge from u to v when v may follow u, which the condition decides by inspection. An admissible array is then exactly a walk, so with M the adjacency matrix of that digraph on $S = 6561$ vertices, and u, w the vectors recording which windows may start and end an array,

$$a(n) \ = \ u^\top M^{\,g(n)} w$$

for the linear function g that the entry's shape supplies. In particular a is C-finite: it satisfies some monic integer linear recurrence, though not necessarily a short one.

3 A closed form is a recurrence

Lemma 1. *Let* $f(m) = \sum_j p_j(m)\lambda_j^m$ *with the* λ_j *distinct and the* p_j *polynomials, and put* $q(x) = \prod_j (x - \lambda_j)^{\deg p_j + 1}$*. Then* q *applied to the shift operator annihilates* f*: writing* $q(x) = x^r - \sum_{i=1}^r c_i x^{r-i}$*,*

$$f(m) \;=\; \sum_{i=1}^{r} c_i\, f(m-i) \qquad \textit{for every } m.$$

Proof. The shift operator E acts on $m \mapsto \lambda^m p(m)$ as $(E-\lambda)\big(\lambda^m p(m)\big) = \lambda^{m+1}\big(p(m+1) - p(m)\big)$, and $p(m+1) - p(m)$ has degree one less than p. So $(E-\lambda)^{\deg p + 1}$ kills $\lambda^m p(m)$, and the factors of q commute. □

Here $q(x) = (x - 16)(x - 1)$, of degree 2, and the recurrence it encodes is

$$a(n) \;=\; 17\, a(n-1) - 16\, a(n-2). \tag{1}$$

4 The proof

Lemma 2. *With* q *as above,* a *satisfies* (1) *for all large* n *if and only if* $u^\top M^j q(M) w = 0$ *for every* $j \geq 0$*, and it is enough to check* $j < S$*.*

Proof. Substituting the walk expression, the residual $a(n) - \sum_i c_i a(n-i)$ equals $u^\top M^{g(n)-r} q(M) w$ up to the fixed linear reparametrisation g. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the values $u^\top M^j q(M) w$ satisfy the monic recurrence given by the characteristic polynomial of M; S consecutive zeros therefore force all later ones, and the last nonzero value pins the threshold exactly. □

Theorem 3. $a(n) = f(n)$ *for every* $n \geq 1$*.*

Proof. The vector $w' = q(M) w$ was formed by 2 matrix–vector products in exact integer arithmetic, and $u^\top M^j w'$ evaluated for $j = 0, 1, \ldots,$ again exactly. Those integers vanish from the point corresponding to $n = 3$ onwards, and the run was continued until the working vector was identically zero, which settles every larger j at once. So a satisfies (1) for every $n > 2$. By Lemma 1 f satisfies (1) identically. Both were then evaluated in exact arithmetic at the 2 consecutive indices $n = 3, \ldots, 4$ and agree at each. A monic recurrence of order 2 determines $a(m)$ from $a(m-1), \ldots, a(m-2)$, and for every $m \geq 5$ both $m > 2$ and $m - 2 \geq 3$ hold, so induction on m gives $a(m) = f(m)$ for every $m \geq 3$. Below $n = 3$ the recurrence argument gives nothing, since (1) is only established for a from $n = 3$ on. The remaining indices $n = 1, \ldots, 2$ were therefore checked one at a time, directly against the walk count in exact integer arithmetic, and agree. □

The range is tight in the sense that was checked: the closed form holds from the entry's first term onwards.

5 Verification

Four checks, each independent of the algebra above.

First, the walk model reproduces every one of the 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the model is built by reading the entry's English, and a misreading gives a different digraph and different counts.

Second, the closed form was evaluated in exact rational arithmetic — not floating point — at every published term from $n = 1$ on, and agrees with the entry's own DATA at each.

Third, the annihilation test of Lemma 2 was carried out exactly, and the model and the closed form were then compared directly at sixty consecutive indices beyond $n = 1$, far past where the proof already settles the matter.

Fourth, the closed form was deliberately perturbed — by a constant and by a term linear in n — and each perturbed form was rejected by the same comparison, so the test is not one that anything would pass.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A209533.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.